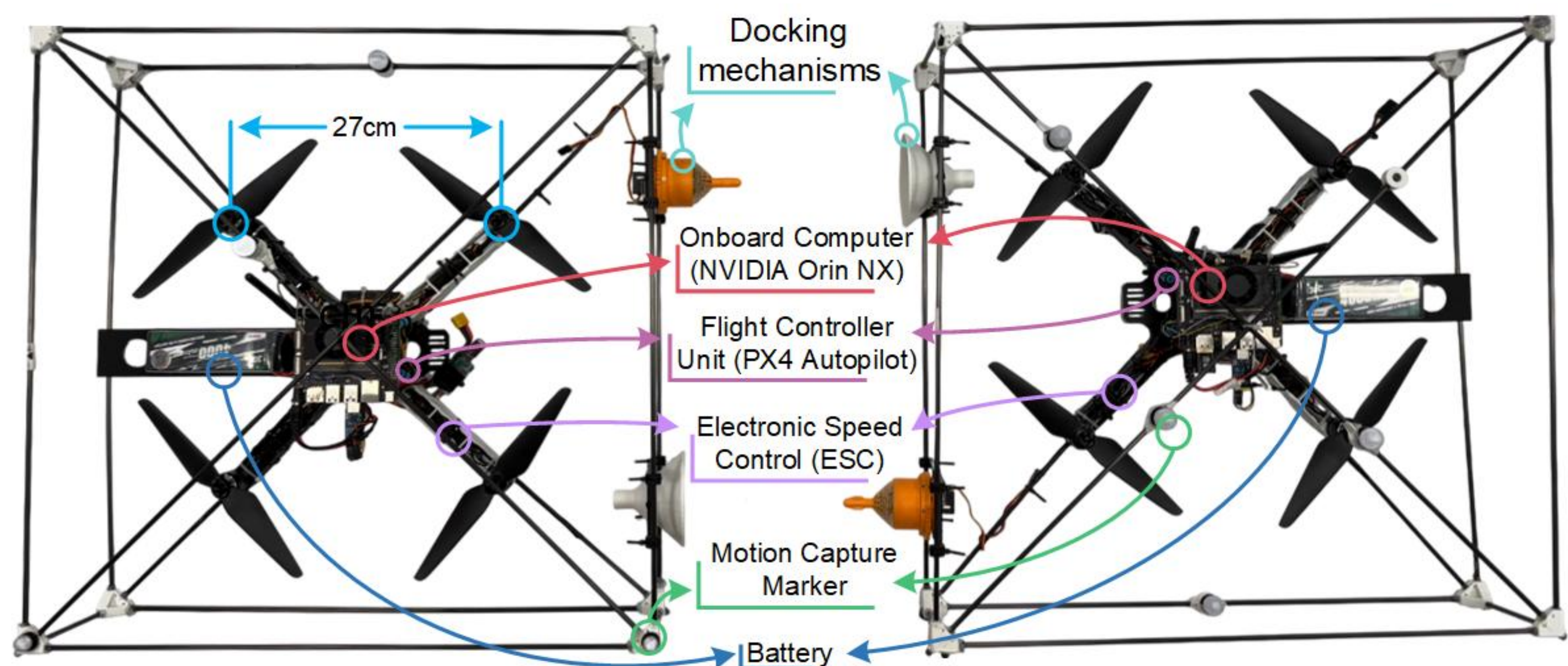


MARS²: Agile and Robust Flight Control of Modular Aerial Robot Systems

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Motivations

- ❑ **MARS capabilities:** adaptive reconfiguration for diverse tasks and fault-tolerant operation.
- ❑ Can we integrate docking, locking, and separation mechanisms into one system?
- ❑ Existing methods rely on manually designed rule-based control allocation.
- ❑ **Objective:** control MARS like a conventional quadrotor to enable use of existing algorithms.



Modular Aerial Robot Systems (MARS)

Methodology

- ❑ **Design:** A mechanism inspired by the dragonfly mating process, enabling **passive docking**, **detection-free locking**, and **active separation** for MARS.
- ❑ **Equivalent model:** Equivalent modeling unifies diverse MARS configurations into a virtual quadrotor, enabling direct use of standard controllers like MPC.
- ❑ **Path-planning and control:** using a model-based controller for diverse demos, including agile trajectory tracking, collision avoidance, and collaborative object transport.

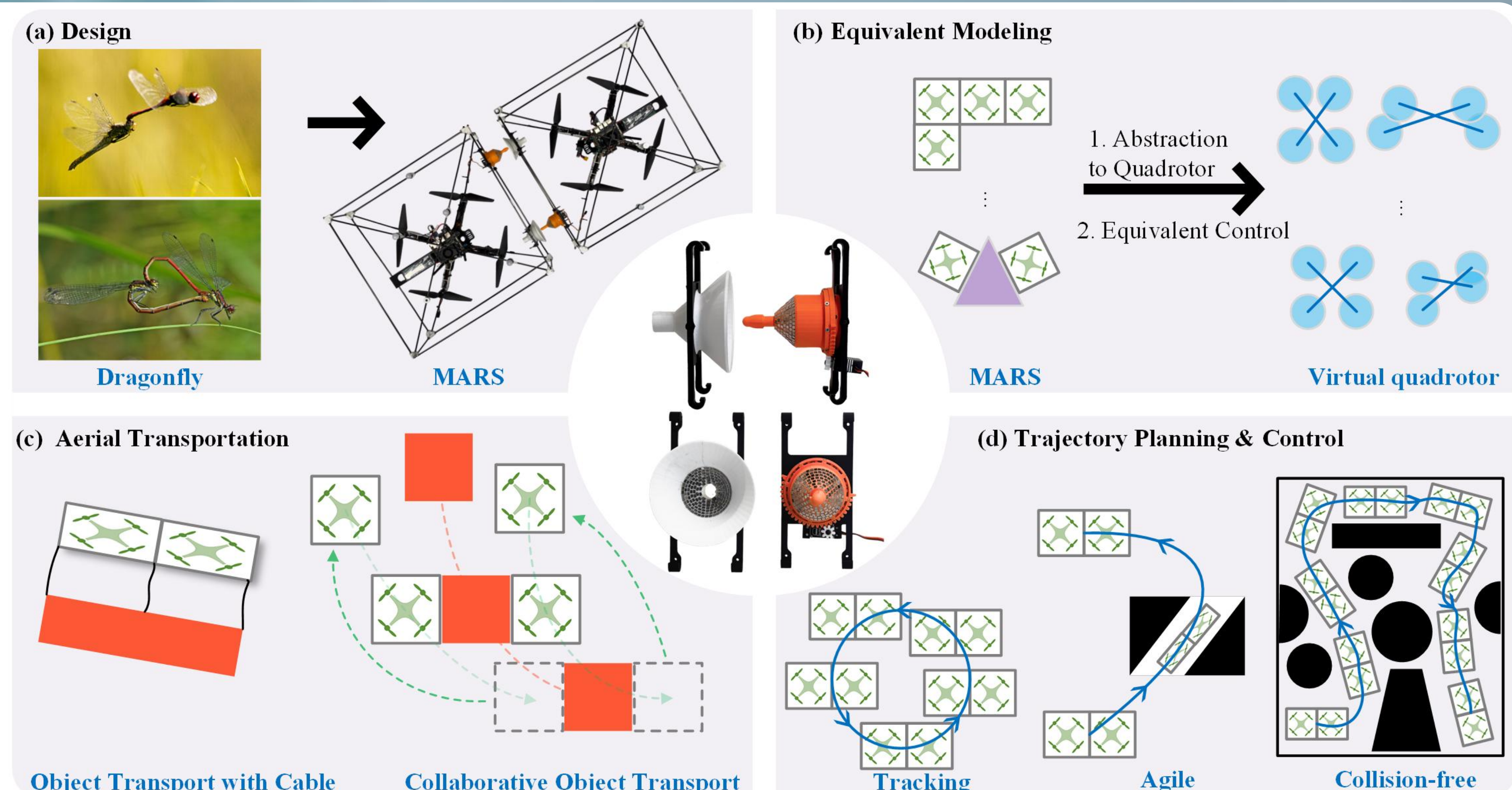
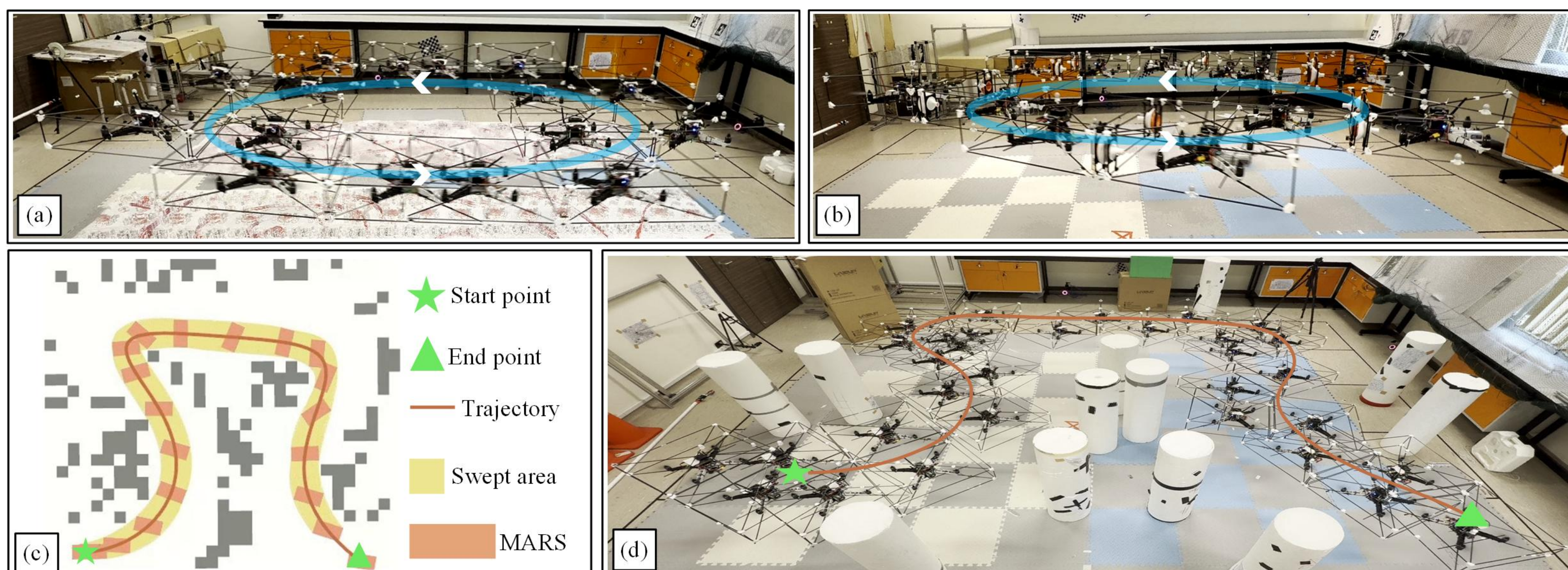
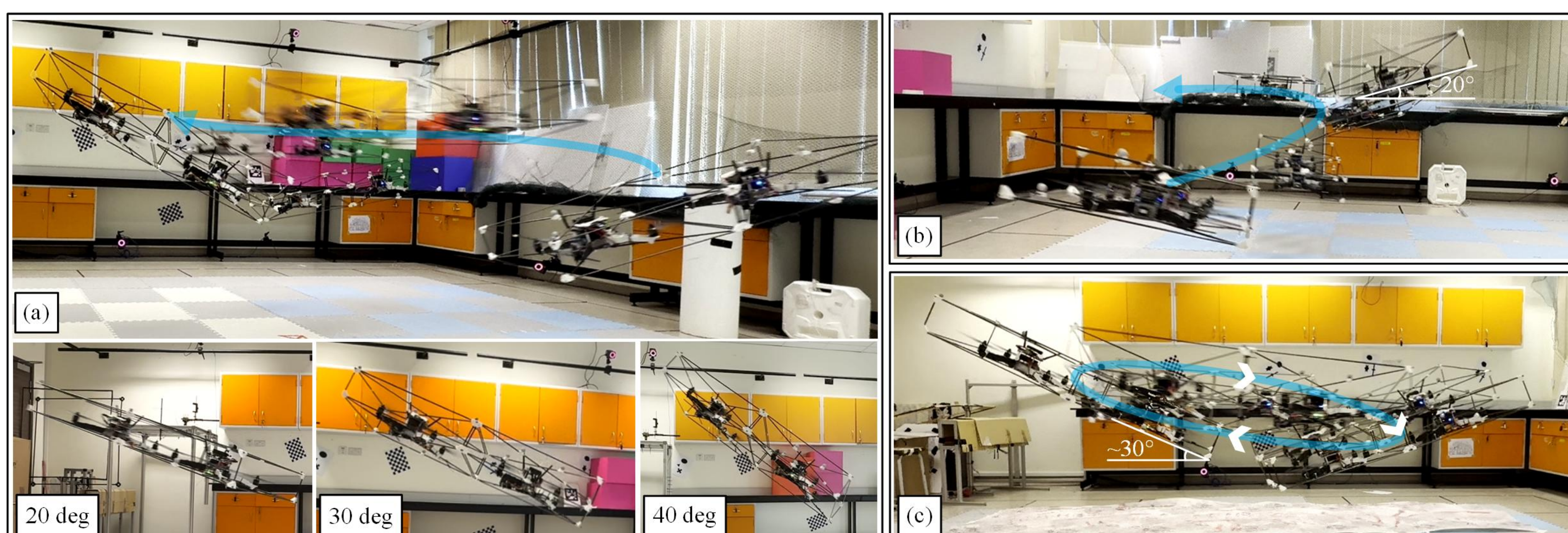


Illustration of the proposed MARS-Dragonfly strategy

Experiment Result



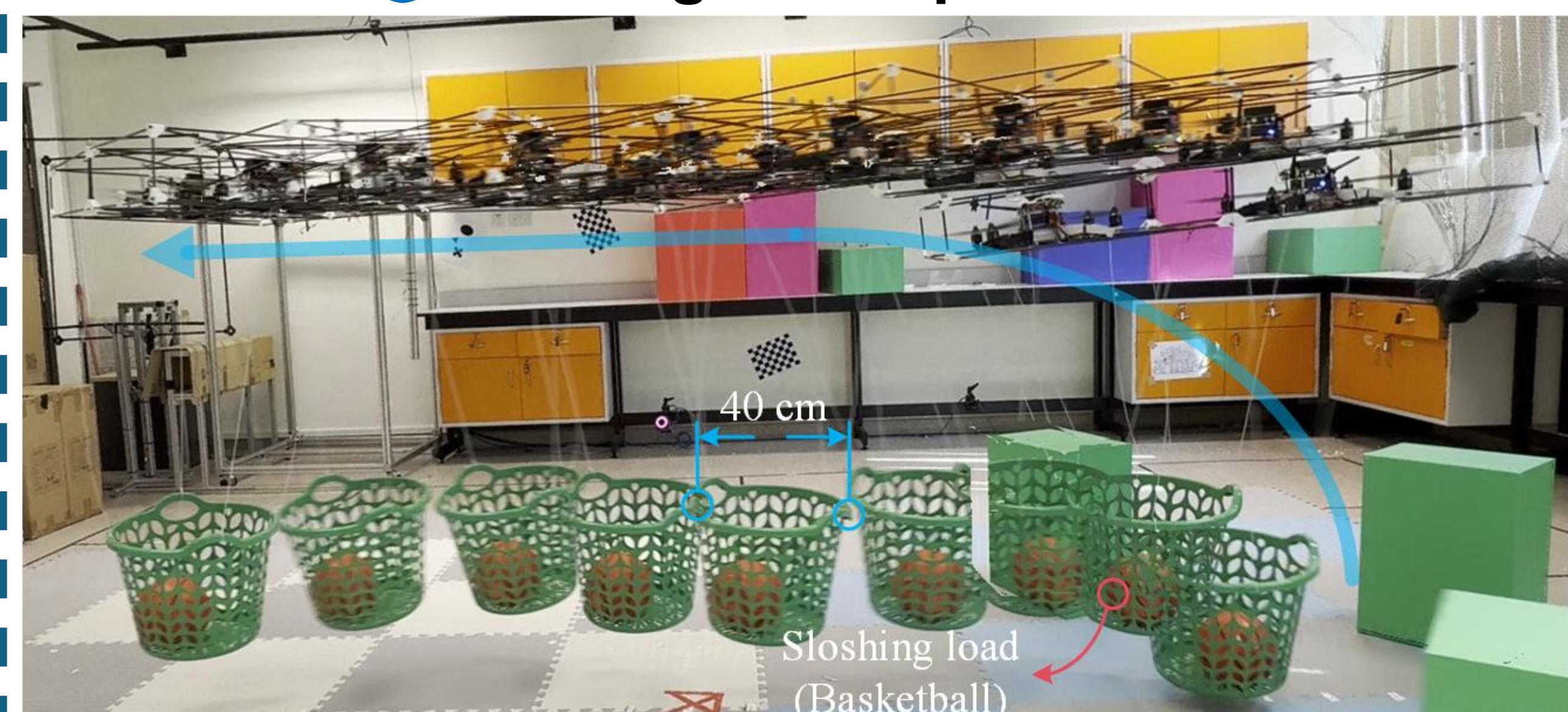
① Trajectory tracking



② Agile trajectory tracking



③ Docking and separation



④ Object transportation

Interested?
watch the video, and learn more.

